

Report

Aditya Singh Yadav(2022039)

Section A

Aditya Singh Yadav
(2022039)

Page No.
 Date

--	--	--

Section - A

(a) let P = profit increase, D = event where company issued dividend

companies that issued dividend last year have 10% profit increase, $P(P|D) = 0.1$

$P(P|ND) = 0$ as company who did not issue dividend has 0% profit.

spread = 36% which is a normal (uniform) distribution.

standard deviation $\sigma = 36$

Also given that $P(D) = 0.8$

To find: $P(D|P=4\%) = \frac{P(4\%|D) \cdot P(D)}{P(P=4\%)}$

$P(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

PDF of normal distribution

$P(P=4\%|D) = \frac{1}{\sqrt{2\pi} \cdot 36} e^{-\frac{(4-10)^2}{2 \cdot (36)^2}}$

$= \frac{1}{\sqrt{2\pi} \cdot 36} e^{-\frac{(4-10)^2}{2 \cdot (36)^2}}$

$= \boxed{0.0108}$

$P(P=4\%|ND) = \frac{1}{\sqrt{2\pi} \cdot 36} e^{-\frac{(4-0)^2}{2 \cdot (36)^2}}$

$= 0.011 \times 0.992$

$= \boxed{0.0109}$

$$\begin{aligned}
 P(P=4\%) &= P(P=4\%|D) \cdot P(D) + P(P=4\%|\neg D) \cdot P(\neg D) \\
 &= 0.8 \times 0.0108 + 0.2 \times 0.0109 \\
 &= 0.01082
 \end{aligned}$$

$$\begin{aligned}
 P(D|P=4\%) &= \frac{P(P=4\%|D) \times P(D)}{P(P=4\%)} \\
 &= \frac{0.0108 \times 0.8}{0.01082} \\
 &= 0.7985
 \end{aligned}$$

$\therefore$ Likelihood that a company with 4% profit increase will issue dividend = 79.85%

$$\begin{aligned}
 (b) \quad P(Y = \text{Attending ML class}) &= \frac{7}{12} \\
 P(N_Y) &= \frac{5}{12}
 \end{aligned}$$

$$\begin{aligned}
 E(Y) &= -P(Y) \log_2(P(Y)) - P(N_Y) \log_2(P(N_Y)) \\
 &= -0.58(-0.778) - 0.416(-1.263) \\
 &= 0.9767
 \end{aligned}$$

• 1st for Attribute (had proper sleep)

$$\begin{aligned}
 P(Y|X_1 = \text{had proper sleep}) &= 1 \\
 P(N_Y|X_1) &= 0
 \end{aligned}$$

$$P(Y|N_{X_1}) = 1/6 \Rightarrow P(N_Y|N_{X_1}) = 5/6$$

$$\begin{aligned}
 E[Y|X_1] &= -P(Y|X_1) \log_2(P(Y|X_1)) - P(N_Y|X_1) \log_2(P(N_Y|X_1)) \\
 &= -1 \log_2(1) - 0 \cdot \log_2(0) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 E(Y/X_1) &= -P(Y/X_1) \log_2(P(Y/X_1)) - \\
 &\quad P(NY/X_1) \log_2(P(NY/X_1)) \\
 &= -\frac{1}{6} \log_2(1/6) - \frac{5}{6} \log_2(5/6) \\
 &= 2.58 + 0.83 (0.26) \\
 &= 0.65
 \end{aligned}$$

weighted average of entropy due to split by $X_1 = \text{Had proper sleep.}$

$$\begin{aligned}
 E(X_1) &= P(X_1) E(Y/X_1) + P(NX_1) E(Y/NX_1) \\
 &= \frac{1}{2} \times 0 + \frac{1}{2} \times 0.65 \\
 &= 0.325
 \end{aligned}$$

$$I(Y, X_1) = E(Y) - E(X_1) = 0.65$$

• 1st for Attribute (class time)

$$P(Y/X_2 = \text{Morning}) = 3/4$$

$$P(NY/X_2) = 1/4$$

$$\begin{aligned}
 E(X_2) &= -\frac{3}{4} \log_2(3/4) - \frac{1}{4} \log_2(1/4) \\
 &= 0.811 + 0.5
 \end{aligned}$$

$$E(X_2) = 0.811$$

$$P(Y/X_2 = \text{Noon}) = 1/2$$

$$P(NY/X_2 = \text{Noon}) = 1/2$$

$$E(X_2 = \text{Noon}) = -\frac{1}{2} \log_2(1/2) - \frac{1}{2} \log_2(1/2)$$

$$= 1$$

$$P(Y/X_2 = \text{Afternoon}) = 1/2; P(NY/X_2 = \text{Afternoon}) = 1/2$$

$$E(X_2 = \text{Afternoon}) = 1$$

$$E(X_2 - \text{class time}) = \frac{1}{3} \times 0.811 + \frac{1}{3} + \frac{1}{3}$$

$$= 0.937$$

$$I_G(Y, X_2) = E(Y) - E(X_2)$$

$$= 0.039$$

• I_G for weather Attribute,

$$P(Y/X_3 = \text{Cool}) = \frac{4}{5}, P(NY/X_3 = \text{Cool}) = 1/5$$

$$E(\text{Cool}) = -\frac{4}{5} \log_2\left(\frac{4}{5}\right) - \frac{1}{5} \log_2\left(\frac{1}{5}\right)$$

$$= 0.257 + 0.469$$

$$= 0.72$$

$$P(Y/X_3 = \text{Raining}) = \frac{0}{2} = 0$$

$$P(NY/\text{Raining}) = 1$$

$$\therefore E(\text{Raining}) = 0$$

$$P(Y/X_3 = \text{Hot}) = \frac{3}{5}, P(NY/X_3 = \text{Hot}) = 2/5$$

$$E(\text{Hot}) = -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right)$$

$$= 0.133 + 0.159 = 0.292$$

$$E(\text{weather}) = \frac{5}{12} \times 0.72 + \frac{2}{12} \times 0 + \frac{5}{12} \times 0.292$$

$$= 0.42$$

$$I_G(Y, X_3) = E(Y) - E(X_3)$$

$$= 0.97 - 0.42$$

$$= 0.55$$

$\therefore$ I_G of proper sleep is highest so split will be done on the basis of proper sleep.

$$E(\text{proper sleep}) = 0.65$$

[NO]

Proper sleep

[yes]
yes

$$E(\text{proper sleep}) = 0$$

Let y' be the event of attending lecture without proper sleep.

$$P(y') = 1/6$$

→ Splitting on basis attribute → classmate

$$P(y' / X_2 = \text{Morning}) = 1/2$$

$$\therefore E(\text{Morning}) = 1$$

$$P(y' / X_2 = \text{Noon}) = 0$$

$$E(\text{Noon}) = 0$$

$$P(y' / X_2 = \text{Afternoon}) = 0$$

$$E(\text{Afternoon}) = 0$$

$$E(\text{classmate}) = \frac{1}{3} \times 1 + \frac{1}{3} \times 0 + \frac{1}{3} \times 0 = \frac{1}{3}$$

$$IG = 0.65 - 0.33 = 0.32$$

Splitting on basis of weather.

$$P(y' / X_3 = \text{Cool}) = 1/2$$

$$E(\text{Cool}) = 1$$

$$P(y' / X_3 = \text{Rainy}) = 0$$

$$E(\text{Rainy}) = 0$$

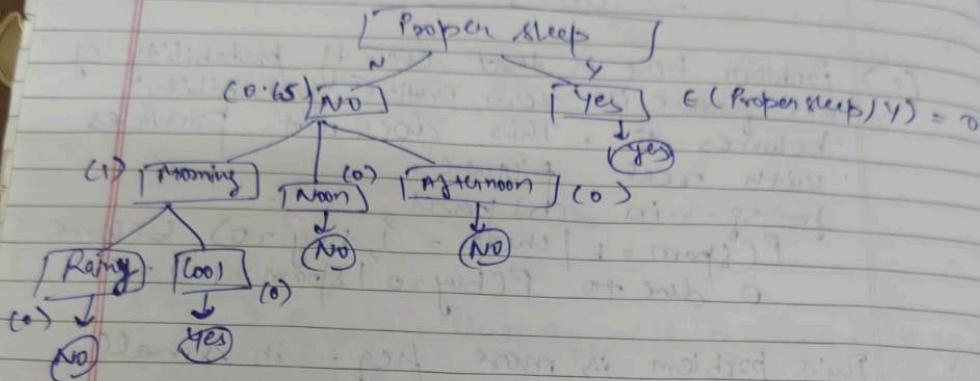
$$P(y' / X_3 = \text{Hot}) = 0$$

$$E(\text{Hot}) = 0$$

$$E(\text{weather}) = \frac{5}{12} \times 1 + \frac{2}{12} \times 0 + \frac{5}{12} \times 0 = 5/12$$

$$1.01 = 0.65 + 5/12 = 0.233$$

1. On due to class time is high, so next split will be on basis of class-time.



(D)

$$(a) P(\text{buy} = 1 | \text{spam} = 1) = \frac{2}{2+1} = \frac{2}{3}$$

$$P(\text{buy} = 0 | \text{spam} = 1) = \frac{1}{2+1} = \frac{1}{3}$$

$$P(\text{cheap} = 1 | \text{spam} = 1) = \frac{1}{2}$$

$$P(\text{cheap} = 0 | \text{spam} = 1) = \frac{1}{2}$$

$$P(\text{buy} = 1 | \text{spam} = 0) = \frac{1}{2}$$

$$P(\text{buy} = 0 | \text{spam} = 0) = \frac{1}{2}$$

$$P(\text{cheap} = 1 | \text{spam} = 0) = \frac{1}{2}$$

$$P(\text{cheap} = 0 | \text{spam} = 0) = \frac{1}{2}$$

$$(b) P(\text{spam} = 1 | \text{buy} = 0, \text{cheap} = 1)$$

$$= P(\text{buy} = 0 | \text{spam} = 1) \cdot P(\text{cheap} = 1 | \text{spam} = 1)$$

$$P(\text{spam} = 1)$$

$$= 0 \times \frac{1}{2} \times \frac{1}{2}$$

$$= 0$$

$$P(\text{spam} = 0 | \text{buy} = 0, \text{cheap} = 1) = P(\text{buy} = 0 | \text{spam} = 0) \cdot P(\text{cheap} = 1 | \text{spam} = 0) \cdot P(\text{spam} = 0)$$

$$= 0.5 \times 0.5 \times 0.5$$

$$= 0.125$$

(c) Problem here is that even if probability of one feature is zero entire probability becomes 0. This does not provides with real picture.

for eg. in (b) part $P(\text{spam} = 1 | \text{cheap} = 1, \text{buy} = 0)$ become 0 due to $P(\text{buy} = 0 | \text{spam} = 1) = 0$

This problem is more freq. in small dataset.

Solutions are Laplace smoothing & m-estimation.

① Laplace smoothing add a count of 1 to all features

$$P(A_i | C) = \frac{N_{ic} + 1}{N_c + C}$$

② M-estimate add a value m instead of adding 1.

$$P(A_i | C) = \frac{N_{ic} + M p}{N_c + M}$$

(c)

let w_t be the weight vector after every update.

update rule being followed is,

$$w_{t+1} = w_t + \eta x_i; \quad w^*: \text{correct weight vector} \quad (1)$$

After each update $w_t \cdot w^*$ increase by η at least.

After t iteration,

$$w_t \cdot w^* \geq t\eta \quad (2)$$

from (1)

$$\|w_{t+1}\|^2 = \|w_t + \eta x_i\|^2 = \|w_t\|^2 + 2\eta (w_t \cdot x_i) + \eta^2 \|x_i\|^2$$

$$= \|w_t\|^2 + 2\eta (w_t \cdot x_i) + 1$$

↓

($\|x_i\|^2 = 1$ as euclidean length is same).

$$\Rightarrow \|w_t\|^2 \leq t \quad (3)$$

$$\|w_t\| \leq \sqrt{t}$$

Applying Cauchy - Schwarz inequality

$$\|u\|^2 \|v\|^2 \geq [u \cdot v]^2$$

$$\Rightarrow w_t \cdot w^* \leq \|w_t\| \|w^*\|$$

$$t\eta \leq w_t \cdot w^* \leq \|w_t\| \|w^*\| \leq \sqrt{t} \|w^*\|$$

2) $t \leq \sqrt{t} \|w^*\|$
 $\Rightarrow t \leq \frac{\|w^*\|^2}{\gamma^2}$

As $\|w^*\|$ is const. assuming it as $2\sqrt{2}$

$\Rightarrow \boxed{t \leq \frac{8}{\gamma^2}}$

if margin is $(1-\epsilon)\gamma$ instead of $\frac{\gamma}{2}$

~~Ans~~ Answer $\Rightarrow \frac{8}{(1-\epsilon)^2 \gamma^2}$

Section B

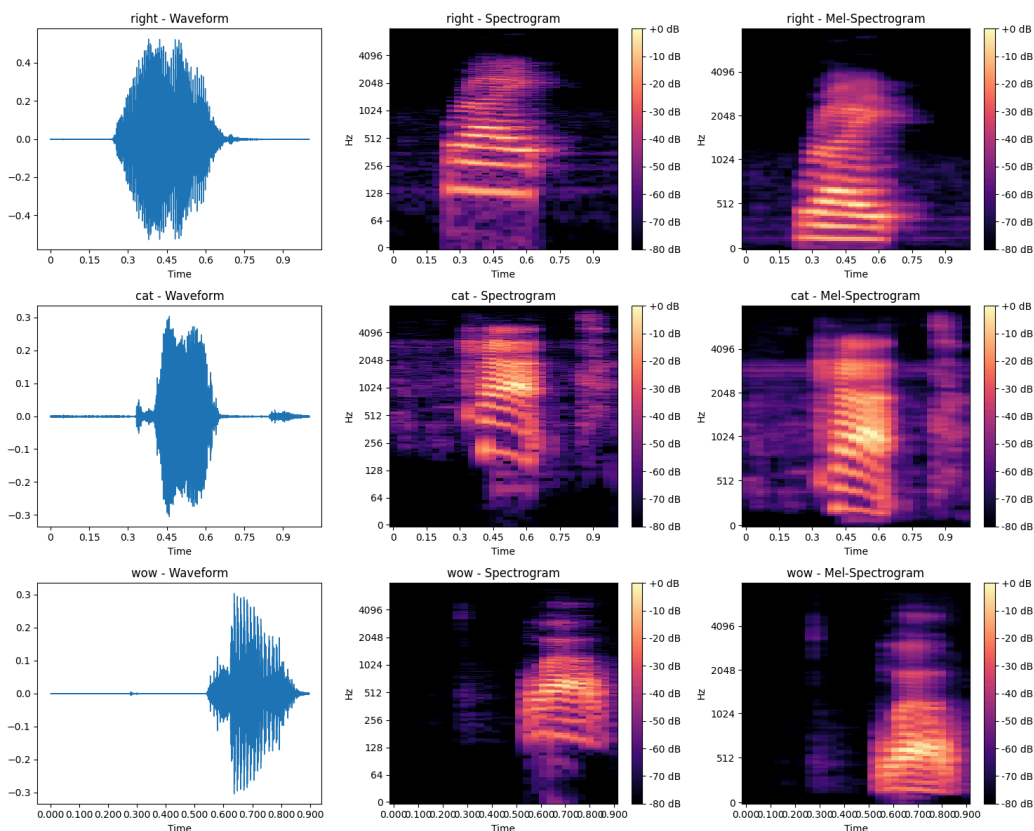
1. (4 points) Perform extensive exploratory data analysis (EDA) by performing the below-given sub-tasks:

(a) Find the statistical summary of the amplitude values and duration distribution of the audio files for each class. Write your observations.

The analysis indicates that the audio data across various classes is generally well-normalized and exhibits consistency in both amplitude and duration.

However, the background noise category deviates from this pattern, showing significant differences in duration and amplitude variability. This divergence is anticipated, as background noise typically contrasts with the shorter, more uniform clips associated with other categories, such as "bird" or "dog," which tend to have more consistent lengths.

(b) Plot graphical representations such as waveform plots, spectrograms, Mel-spectrograms, etc for randomly selected 3 audio files from randomly selected 3 classes out of 35 classes. Write your observations.



Waveforms:

- **Right:** Symmetric and dense, indicating a burst of sound energy with rapid decay.
- **Cat:** Less symmetric, energy concentrated in the middle, tapering off gradually.
- **Wow:** Distinctive energy burst in the second half, suggesting a longer sound event.

Spectrograms:

- **Right:** Energy primarily in low frequencies (<512 Hz) with some high-frequency components.
- **Cat:** Energy across low and mid-range (512-2048 Hz), with gradual decay.
- **Wow:** Concentrated in mid-frequencies (128-1024 Hz) toward the latter part.

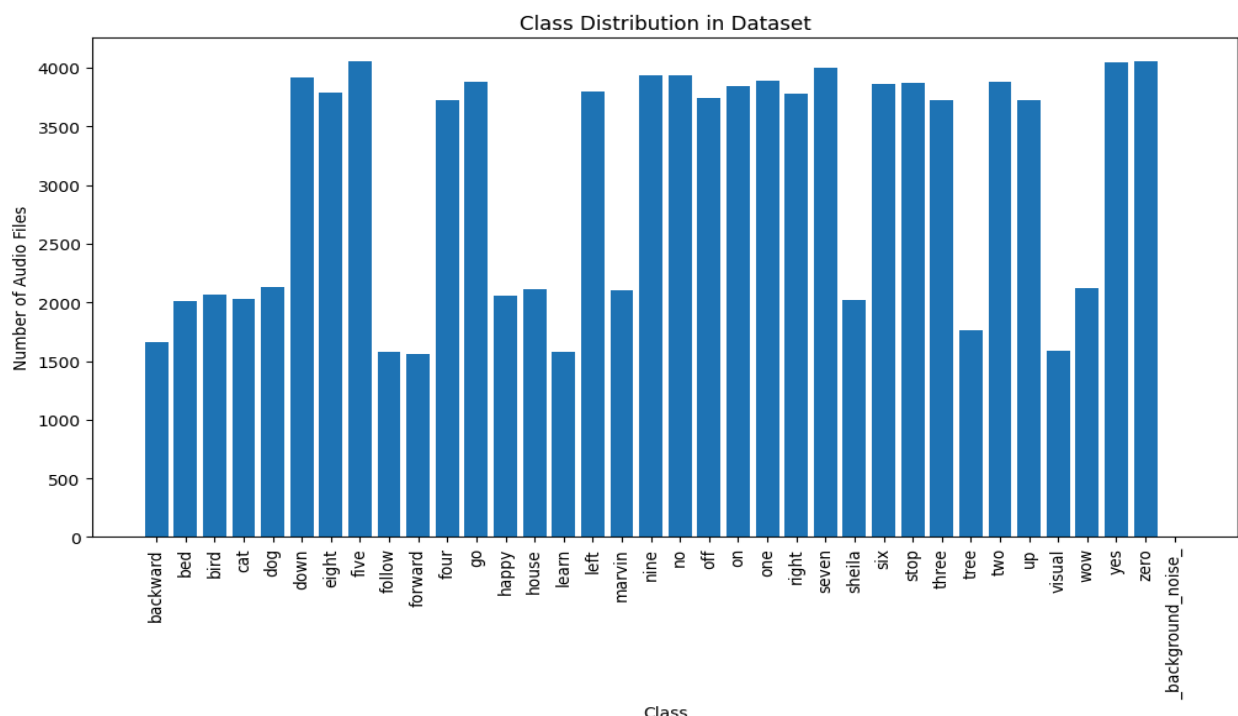
Mel-Spectrograms:

- **Right:** Dense low-frequency concentration with well-defined horizontal energy bands.
- **Cat:** More evenly spread energy across frequencies, appearing more dispersed.
- **Wow:** Strong mid to low frequency presence, especially distinct at the end.

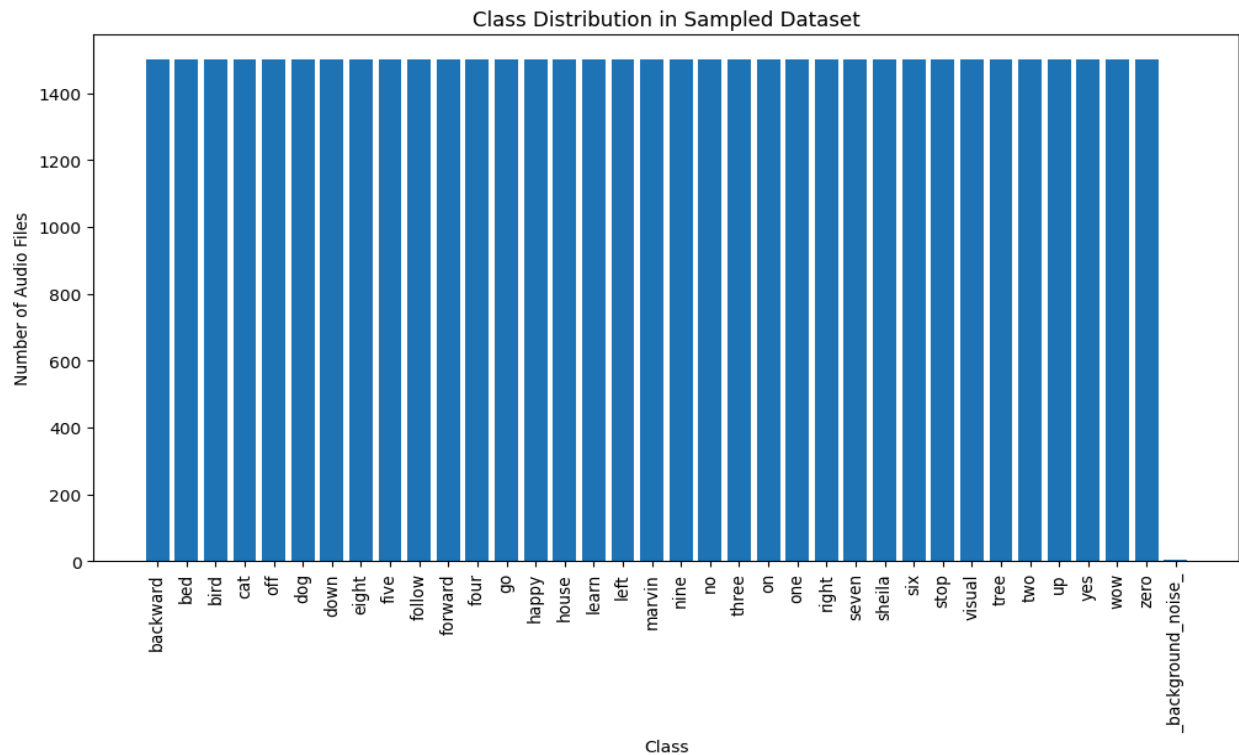
Comparison:

- **Right:** More symmetric waveform and spectrogram, dominant in low frequencies.
- **Cat:** Even energy distribution across a wider frequency range, resulting in a dispersed appearance.
- **Wow:** Concentrated energy in the middle with distinct mid-frequency characteristics.

(c) Investigate any class imbalances in the dataset, and if present, handle them through techniques like oversampling, undersampling, or data augmentation.



Yes there is class imbalance so i have done undersampling of 1500 rows per class,
So its like 35*1500 rows total and also done outlier removals based on durations using IQR .



2. (3 points) Perform Feature Extraction: Extract relevant features from the audio data, such as MFCCs, chroma features, spectral features, and features extracted using filters like bandpass filter, etc.

Ans)

1. ****MFCCs****:
 - Mean
 - Variance
2. ****Chroma Features****:
 - Mean
 - Variance
3. ****Spectral Features****:
 - Spectral Centroid (Mean)
 - Spectral Bandwidth (Mean)
 - Spectral Contrast (Mean)
4. ****Zero Crossing Rate**** (Mean)

5. ****Short-Time Energy**** (Mean)

6. ****Filtered MFCCs****:

- Mean
- Variance

These features collectively capture important aspects of the audio signals for classification.

3) (3 points) Model Selection and Implementation:

a) You are allowed to use only any one or ensemble of Naive Bayes, Decision Tree, Random Forest, and Perceptron models, all implemented from scratch. Train and test these models on the prepared dataset using an 80:20 train-test split.

I have used LightGBM classifier which is an ensemble learning model that uses decision trees as its base learners. Specifically, it employs a gradient boosting framework, which builds trees sequentially to minimize the loss function.

(b) Evaluate the selected models using accuracy as the metric. Document the results and compare the performance of different models. (Note: The use of libraries like scikit-learn, PyTorch, etc., is strictly prohibited for this task. All steps must be implemented from scratch.)

Best Performer: The ensemble method stands out as the most effective model, achieving the highest accuracy at 63.41%. This suggests that leveraging multiple models can lead to better predictions.

Moderate Performer: Random Forest is also a strong contender, providing better accuracy than individual classifiers.

Underperformers: Naive Bayes, Perceptron, and Decision Tree struggle, with accuracies below 36%, highlighting their limitations in this particular task.

Section C

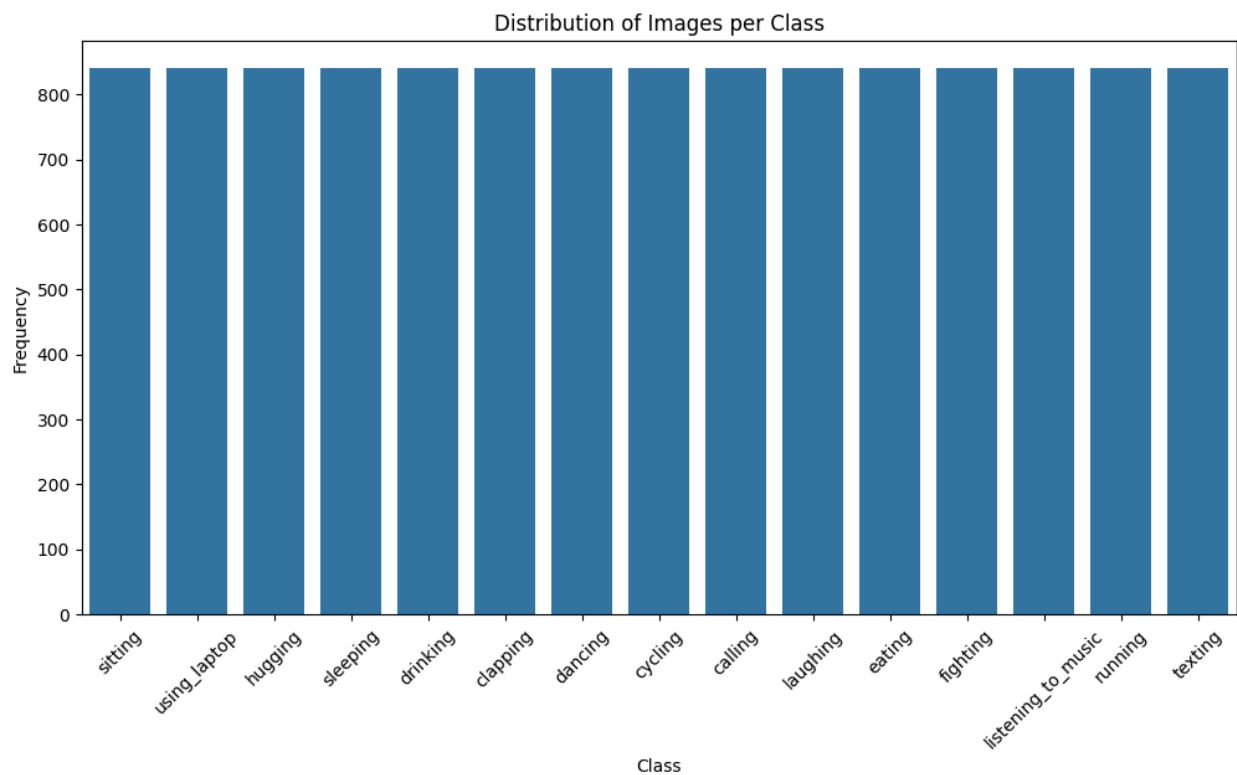
1. (3 marks) Perform EDA:

(a) Provide an overview of the dataset, including the number of images per class, the distribution of image sizes, and any other relevant statistics. Write your observations.

Each class contain 840 images which is equally balanced. The distribution of image sizes is given below:

	weight	height
count	200.000000	200.000000
mean	256.510000	196.805000
std	46.817189	37.800281
min	84.000000	84.000000
25%	248.750000	179.000000
50%	275.000000	183.000000
75%	276.000000	194.500000
max	385.000000	300.000000

(b) Create visualizations to represent the distribution of classes, and display a few sample images from each class. Write your observations.



Each class is equally distributed with 840 image samples

Class: sitting



Class: using_laptop



Class: hugging



Class: sleeping



Class: drinking



Class: clapping



Class: dancing



Class: cycling



Class: calling



Class: laughing



Class: eating



Class: fighting



Class: listening_to_music



Class: running



Class: texting

Class: sitting



Class: using_laptop



Class: hugging



Class: sleeping



Class: drinking



Class: clapping



Class: dancing



Class: cycling



Class: calling



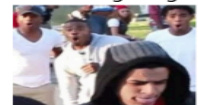
Class: laughing



Class: eating



Class: fighting



Class: listening_to_music



Class: running



Class: texting

(c) Investigate the class distribution and discuss whether there are any class imbalances. If imbalances are present, propose strategies to address them, such as data augmentation or resampling techniques.

There is no class imbalance in the dataset but I have done outlier analysis and removal for better model prediction.

2. (3 marks) Perform Feature Extraction: Extract relevant features from the images such as Histogram of Oriented Gradients (HOG), color histograms, or other traditional feature extraction methods. Write your observations.

I have used Histogram of Oriented Gradients (HOG), color histograms and The Gray Level Co-occurrence Matrix (GLCM) as the combination of these features were giving best results than other combinations.

3. (2 marks) Model Selection and Implementation:

(a) You are allowed to use only any one or ensemble of Naive Bayes, Decision Tree, Random Forest, and Perceptron models. Train and test these models on the prepared dataset using an 80:20 train-test split.

I have used the LightGBM classifier which is an ensemble learning model that uses decision trees as its base learners. Specifically, it employs a gradient boosting framework, which builds trees sequentially to minimize the loss function.

(b) Evaluate the selected models using accuracy as the metric. Document the results and compare the performance of different models.

LightGBM significantly outperforms the other models with accuracy of 44.66, suggesting that it's more suited for this particular dataset.

Random Forest also shows good results, benefiting from its ability to handle overfitting compared to a simple decision tree.

The traditional models (Perceptron, Decision Tree, and Naive Bayes) performed poorly, indicating that they might not be ideal for the given data characteristics.

Overall, the results highlight the importance of model selection based on dataset complexity and characteristics. Using more advanced models like LightGBM can lead to better performance.